# Mateo Mangialomini

M.Sc.A. Candidate in ML/AI — Polytechnique Montréal

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## Research Experience

**Graduate Researcher — Polytechnique Montréal × Atelier7Hz** — Montréal, Canada
*Multi-teacher knowledge distillation for edge sound event detection* — *Jan 2026 – Present*

- Built a teacher ensemble surpassing state-of-the-art on FSD50K (+**0.4 pts mAP**) with **93.7% fewer parameters**.
- Authored a novel confidence-weighted multi-teacher KD method: distilled BEATs, PaSST, AST and MobileNetV3 into an EfficientAT student reaching **67.05 mAP**, above every teacher (all <**65.3**). First-author paper in preparation.
- Mapped the accuracy–compute Pareto front across pruning, INT8 quantization, and combined schemes to select operating points for Raspberry Pi 4B deployment.
- Wrote custom C++/CUDA kernels for mel-spectrogram computation with shared-memory 2D tiling, achieving a **2× speedup** over torchaudio on an NVIDIA RTX 5090 during training.
- Rewrote the inherited training pipeline (PyTorch, Hydra, Ray Tune/Optuna, ONNX) for scene classification and flyover detection, removing unnecessary CPU–GPU synchronization calls to speed up training by **13%**.

## Education

**M.Sc.A. in ML/AI — Polytechnique Montréal** — Montréal, Canada
*Thesis: Multi-teacher knowledge distillation for edge audio classification* — *Sep 2025 – Apr 2027*

- Coursework: Deep Learning (A. Courville, Mila) A*, Probabilistic ML (C. Pal) A*, Machine Learning A.

**Arts et Métiers ParisTech — Top French Engineering School — Class Rank: 2 /1149** — France
*Mechanical, Electrical & Industrial Engineering — Double degree with Polytechnique Montréal* — *2023 – 2027*

- Coursework: Solid Mechanics, Fluid Mechanics, Vibration & Dynamics, Computational Engineering, AI; hands-on machining, casting and welding.

**CPGE (Preparatory Classes) — Lycée Masséna** — Nice, France
*Intensive mathematics & physics for Grandes Écoles entrance exams* — *2021 – 2023*

**Baccalauréat, High Honors — Lycée International de Valbonne (CIV)** — Valbonne, France
*Mathematics & Physics — International curriculum in Italian* — *2021*

## Leadership & Involvement

**PolyAI — Competitions Department (Gestionnaire)** — Montréal, Canada
*Polytechnique Montréal AI student association* — *Mar 2026 – Present*

- Coordinate ML competition training and compete in hackathons; event staff at CodeML, Polytechnique's ML hackathon.

## Professional Experience

**Airbus Helicopters — Prototype Department** — Marignane, France
*Intern — Shop Floor* — *Jun – Jul 2024*

**MANE — Flavor & Fragrance Manufacturer** — Le Bar-sur-Loup, France
*Technical Maintenance Worker* — *Jul – Aug 2023*

## Projects

**ESC-50 Environmental Sound Classification — live demo**
*PyTorch, librosa, FastAPI, Docker, HuggingFace Spaces* — *2026*

- End-to-end deep learning project, self-taught: extended the Salamon & Bello (2017) CNN from 10 to 50 classes maintaining accuracy and deployed it as a FastAPI + Docker endpoint with a real-time web UI.

**Kaggle Competition — 3rd / 39**
*XGBoost, CatBoost, LightGBM* — *2025*

- Predicted E. coli antibiotic resistance from 1M-dimensional sparse genomic features under class imbalance and phylogenetic distribution shift, using an out-of-fold calibrated mixture-of-experts gradient boosting ensemble.

**Hydrogadz — Monaco Energy Boat Challenge 2025** — Monaco
*Student association — electric boat* — *2025*

- Designed the helical gear transmission and fabricated a symmetric NACA-profile aluminium rudder (MAG welding).

**Turbo-Compressor Prototype — R&D Project**
*Ansys VistaCCD, Siemens Star-CCM+, 3D printing* — *2025*

- Designed a centrifugal compressor first stage analytically, validated it via symmetry-reduced CFD (5.4% compression error achieved wrt the simulation), and 3D-printed a functional prototype.

## Technical Skills & Languages

- **Languages:** Python, C++/CUDA, SQL
- **ML & Data:** PyTorch, scikit-learn, XGBoost/LightGBM/CatBoost, NumPy, Pandas, librosa, ONNX, quantization, pruning
- **MLOps & Tooling:** Hydra, Ray Tune, Optuna, Docker, FastAPI, Git, Linux, HuggingFace, MATLAB, Ansys
- **Spoken:** French (native), Italian (C2), English (C1 — TOEIC 990/990), German (B1/B2 — self-taught)